

Practice Quiz: Perception -- How You Construct Your Social Reality

Part A: Multiple Choice

1. Active Inference describes perception as: A) A camera passively recording reality B) An active, constructive process where the brain generates predictions and checks them against sensory input C) A process that only works when you pay close attention D) Something that happens only through your eyes

Answer: B. Perception is top-down prediction tested against bottom-up sensory evidence, not passive reception.

1. You can read a text message full of typos without noticing them because: A) Your phone autocorrects before you see it B) Your brain's generative model predicts the correct words and fills in the errors C) Typos are invisible to the human eye D) You are not actually reading the message

Answer: B. Strong top-down predictions override small sensory discrepancies, making typos invisible.

1. **Prior beliefs** in social perception are: A) Opinions you consciously choose to hold B) Expectations your generative model brings to every interaction, shaped by past experience C) Beliefs that are always wrong D) Predictions that only apply to strangers

Answer: B. Priors are efficient predictions from past experience, but they can become self-fulfilling prophecies.

1. If you believe a classmate dislikes you, you will tend to perceive their neutral facial expressions as hostile. This is because: A) They actually do dislike you B) Your strong prior belief shapes your interpretation of ambiguous sensory evidence C) Neutral expressions are always hostile D) Your eyes are not working correctly

Answer: B. When priors are strong and sensory evidence is ambiguous, your model fills in details that confirm existing beliefs.

1. When a reliable friend unexpectedly cancels plans, the **prediction error** you experience is: A) Meaningless noise B) The most informative signal your brain receives, forcing you to update your model of that friend C) A sign your friendship is over D) Evidence that all friends are unreliable

Answer: B. Prediction errors carry information; larger errors demand larger model updates.

1. Active Inference explains selective **attention** through the concept of: A) Willpower B) Precision weighting -- your brain assigns higher weight to prediction errors it considers reliable and important C) The size of your eyes D) How much sleep you got

Answer: B. Precision determines which prediction errors get processed and which are treated as noise.

1. Anxiety makes you hyper-aware of social threats (sideways glances, awkward pauses) because anxiety: A) Improves your eyesight B) Increases precision weighting on social threat signals, making you process more threat-related prediction errors C) Turns off your generative model D) Makes other people actually more hostile

Answer: B. The world has not changed -- your precision weighting has shifted toward threat detection.

1. Two people at the same party can have genuinely different experiences because: A) One of them is lying about what happened B) Each person runs a different generative model with different priors and precision weightings C) Parties are inherently confusing D) One person was not paying attention

Answer: B. No one perceives objectively -- each person's model constructs a different experience of the same event.

1. First impressions are hard to change because: A) People never change their behavior B) The initial impression becomes a strong prior that shapes all subsequent perception, requiring many contradictory prediction errors to override C) Your brain only processes information once D) First impressions are always accurate

Answer: B. Strong priors dominate until enough accumulated prediction error forces a model revision.

1. Misreading the tone of a text message ("fine.") happens because: A) The sender is always angry B) Digital communication strips away tone and body language, so your priors dominate the interpretation with fewer channels for prediction error C) You cannot read D) Phones distort emotional content

Answer: B. With fewer sensory channels to check predictions, your generative model fills in missing information from its own expectations.

Part B: Short Answer

1. Describe a time when you completely misread a social situation. What prior beliefs led your generative model astray, and what prediction error eventually corrected your perception?
2. Explain how the Active Inference account of perception helps explain why eyewitness testimony can be unreliable, even when witnesses are honest.
3. If anxiety increases precision weighting on threat signals, what practical strategy could you use to reduce this effect? Frame your answer using Active Inference concepts.
4. Give an example of a "self-fulfilling prophecy" in your social life and explain how prior beliefs and selective attention create a feedback loop.
5. Why might it be helpful to assume you are wrong about your first impression of someone? What does Active Inference suggest about how to update social perceptions more accurately?